

ATYRAU, Kazakhstan

WMO: 357000

Lat: 47.1161N

Lon: 51.8838E

Elev: -72

StdP: 14.73

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -1.9	(d) -14.2	(e) 2.5	(f) -6.8	(g) -8.6	(h) 3.5	(i) -1.1	(j) 28.6	(k) 30.1	(l) 26.1	(m) 25.0	(n) 6.3	(o) 50	(p) 0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.4	(c) 99.7	(d) 67.7	(e) 97.0	(f) 67.0	(g) 94.3	(h) 66.0	(i) 72.4	(j) 86.0	(k) 70.5	(l) 86.7	(m) 69.1	(n) 86.7	(o) 8.5	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	103.7	79.0	65.6	94.4	77.5	63.2	86.6	76.7	36.1	85.3	34.4	86.1	33.1	86.9	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.8	(b) 22.9	(c) 19.7	(e) -10.7	(f) 102.8	(g) 9.6	(h) 2.7	(i) -17.6	(j) 104.7	(k) -23.2	(l) 106.3	(m) -28.6	(n) 107.9	(o) -35.6	(p) 109.8		
(a) 26.8	(b) 22.9	(c) 19.7	(e) -11.2	(f) 76.6	(g) 9.5	(h) 2.2	(i) -18.0	(j) 78.3	(k) -23.5	(l) 79.6	(m) -28.8	(n) 80.8	(o) -35.7	(p) 82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	20.6	22.2	36.5	53.1	66.9	76.6	81.1	79.0	65.8	50.5	35.0	25.6		
	DBStd	23.52	10.84	11.97	8.78	8.96	8.46	7.49	5.94	7.50	7.89	8.71	10.19	9.52		
	HDD50	3494	910	778	424	65	2	0	0	0	2	102	454	757		
	HDD65	6573	1375	1198	885	368	78	6	0	3	85	452	901	1222		
	CDD50	3949	0	0	5	159	527	798	964	900	474	118	4	0		
	CDD65	1552	0	0	0	12	137	354	499	438	108	4	0	0		
	CDH74	18230	0	0	0	106	1363	4143	6298	5398	898	24	0	0		
	CDH80	9021	0	0	0	19	522	2036	3332	2833	278	1	0	0		
	WSAvg	8.6	8.9	9.5	9.9	10.0	9.2	8.6	7.8	7.5	7.7	7.7	8.1	8.7		
	PrecAvg	7.2	0.6	0.5	0.6	0.6	1.1	0.7	0.4	0.4	0.4	0.6	0.7	0.7		
(15) Precipitation	PrecMax	13.4	1.3	1.1	1.9	1.8	3.5	1.8	1.6	1.5	1.6	1.9	2.0	1.9		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	2.0	0.4	0.3	0.5	0.4	0.9	0.6	0.4	0.5	0.4	0.5	0.4	0.4		
	PrecStd	2.0	0.4	0.3	0.5	0.4	0.9	0.6	0.4	0.5	0.4	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	45.9	63.1	82.7	94.6	101.4	102.2	102.7	93.5	77.1	58.7	44.7		
		MCWB	37.6	40.5	49.9	59.6	63.2	67.1	68.3	68.1	65.6	56.7	50.3	40.2		
	2%	DB	37.2	40.4	56.8	76.9	90.0	97.7	99.4	99.6	86.6	72.3	54.3	41.7		
		MCWB	35.4	36.3	46.4	57.2	62.0	66.5	68.1	67.6	61.6	55.2	48.6	38.9		
	5%	DB	35.1	37.6	52.1	71.7	86.0	94.5	96.9	96.5	82.6	68.0	51.0	38.9		
		MCWB	33.7	34.7	44.0	54.8	61.0	65.0	67.5	66.8	60.7	54.0	46.4	36.4		
	10%	DB	33.3	35.6	48.4	67.6	81.7	90.9	94.0	93.2	79.2	63.8	47.8	36.3		
		MCWB	31.9	33.3	41.3	53.2	59.8	64.2	66.8	65.9	59.4	51.9	43.8	34.4		
	0.4%	WB	38.3	41.1	51.0	61.4	67.3	72.2	75.8	75.6	69.1	61.4	53.6	41.9		
		MCDB	39.8	44.4	60.0	75.4	80.8	84.5	87.1	84.1	84.8	69.7	56.2	43.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	37.2	47.5	59.2	64.8	70.0	72.7	72.4	65.9	58.1	49.8	39.3		
		MCDB	37.0	39.9	55.7	73.6	80.9	85.8	86.0	86.2	77.5	67.1	52.6	41.6		
	5%	WB	33.9	35.1	44.6	57.0	63.3	68.4	70.9	70.2	63.7	55.9	46.8	36.6		
		MCDB	35.0	36.9	51.1	69.2	80.0	85.7	86.7	88.0	76.0	64.5	50.4	38.3		
	10%	WB	32.1	33.4	42.1	54.5	61.7	67.0	69.4	68.6	61.7	53.6	43.7	34.6		
		MCDB	33.2	35.3	47.4	65.1	77.8	84.6	86.6	87.4	74.8	62.4	47.4	36.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.6	14.7	17.2	17.9	19.1	19.4	20.6	19.3	16.3	11.0	8.6		
		MCDBR	8.4	11.6	19.6	21.5	21.8	22.7	22.3	23.4	22.9	21.1	13.8	8.5		
		MCWBR	7.5	8.6	11.4	9.0	8.2	8.0	8.4	9.1	9.7	10.1	8.9	7.1		
		MCDBR	8.0	10.3	17.9	19.5	18.8	19.0	19.5	20.3	18.6	18.2	12.2	7.5		
	5% WB	MCWBR	7.4	8.1	11.0	8.8	8.0	8.5	9.2	9.9	10.2	11.0	8.8	6.8		
		MCWBR	7.4	8.1	11.0	8.8	8.0	8.5	9.2	9.9	10.2	11.0	8.8	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.358	0.376	0.434	0.496	0.480	0.455	0.455	0.460	0.417	0.398	0.372	0.370		
	taud		2.052	2.006	2.006	1.920	2.030	2.163	2.184	2.139	2.203	2.183	2.166	2.076		
	Ebn at Noon		213	238	244	241	251	259	257	249	249	230	209	191		
	Edh at Noon		31	40	47	56	52	46	45	45	38	34	28	28		
(44) All-Sky Solar Radiation	RadAvg		305	572	1040	1549	1966	2183	2100	1851	1407	855	444	260		
	RadStd		53	100	126	118	143	156	102	87	87	75	56	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+2.39	N/A	-1.31	N/A	N/A	+416	+279
Regional (2 neighbors)	N/A	N/A	N/A	+2.39	N/A	-1.31	N/A	N/A	+416	+279

Nomenclature: See separate page